

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12414502
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Xu; Haishen

Lawn mower and cutting blade

Abstract

A cutting blade is capable of being driven to rotate around a first axis. The cutting blade includes cutting portions comprising cutting edges for cutting and extension portions closer to the first axis relative to the cutting portions. Each of the extension portions comprises an upper surface and a lower surface, at least part of the upper surface is a part of a side surface of a first cone with the first axis as an axis, and at least part of the lower surface is a part of a side surface of a second cone with the first axis as an axis.

Inventors:	Xu; Haishen (Nanjing, CN)
Applicant:	Nanjing Chervon Industry Co., Ltd. (Nanjing, CN)
Family ID:	1000008821762
Assignee:	Nanjing Chervon Industry Co., Ltd. (Nanjing, CN)
Appl. No.:	17/535054
Filed:	November 24, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220183227 A1	Jun. 16, 2022

Foreign Application Priority Data

CN	202011441981.5	Dec. 11, 2020
CN	202011452928.5	Dec. 11, 2020

Publication Classification

Int. Cl.: A01D34/73 (20060101); A01D34/69 (20060101); A01D101/00 (20060101)

U.S. Cl.:

CPC **A01D34/73** (20130101); **A01D34/69** (20130101); A01D2101/00 (20130101)

Field of Classification Search

CPC: A01D (34/69); A01D (34/73)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3910017	12/1974	Thorud	56/295	A01D 34/73
5501068	12/1995	Martz	56/DIG.17	A01D 34/73
6470662	12/2001	Burke et al.	N/A	N/A
2003/0221404	12/2002	Hancock et al.	N/A	N/A
2013/0327007	12/2012	Eavenson, Sr. et al.	N/A	N/A
2014/0182257	12/2013	Von Rentzell	N/A	N/A
2015/0082763	12/2014	Nishihara	56/289	A01D 34/63
2018/0177124	12/2017	Kuriyagawa et al.	N/A	N/A
2020/0100426	12/2019	Kurihara et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
104335755	12/2014	CN	N/A
2615903	12/2016	EP	N/A
2248771	12/1974	FR	N/A
2865610	12/2004	FR	N/A
H0646638	12/1993	JP	N/A
2020/063609	12/2016	WO	N/A
2020/063609	12/2019	WO	N/A

OTHER PUBLICATIONS

EPO, extended European search report issued on European patent application No. 21210918.5, dated May 16, 2022, 8 pages. cited by applicant

Primary Examiner: Rocca; Joseph M

Assistant Examiner: Pezzuto; Robert E

Attorney, Agent or Firm: Greenberg Traurig, LLP

Background/Summary

RELATED APPLICATION INFORMATION

(1) This application claims the benefit under 35 U.S.C. § 119(a) of Chinese Patent Application No. CN 202011441981.5, filed on Dec. 11, 2020, and Chinese Patent Application No. CN

TECHNICAL FIELD

(2) The present disclosure relates to a power tool and, in particular, to a lawn mower and a cutting blade.

BACKGROUND

(3) A lawn mower cuts vegetation through a cutting blade. An ineffective load of the blade is an important factor affecting the time of endurance and mowing performance of the lawn mower. However, due to the design requirements for the strength of the blade, it is often difficult to reduce the ineffective load of the blade, which is not conducive to the optimization of the overall performance of the lawn mower.

SUMMARY

(4) In one example, a cutting blade is capable of being mounted to a lawn mower. When the cutting blade is mounted to the lawn mower, the cutting blade is capable of being driven to rotate around a first axis. The cutting blade includes cutting portions including cutting edges for cutting and extension portions closer to the first axis relative to the cutting portions. Each of the extension portions includes a main body, and the main body constitutes a main part of each of the extension portions. Each of the extension portions is configured to have at least three planes passing through the first axis, and cross-sections of the main body on the at least three planes are substantially the same.

(5) In one example, the main body includes a first section and a second section disposed on two sides of a centerline of the main body, and a vertical distance between each point, on an inner edge of a cross-section perpendicular to the first axis, of at least part of the first section and the first axis is the same.

(6) In one example, the main body includes an upper surface and a lower surface, and the upper surface is a part of the side surface of a first cone with the first axis as an axis.

(7) In one example, the lower surface is a part of the side surface of a second cone with the first axis as an axis.

(8) In one example, the main body includes a centerline, and a distance between the upper surface and the lower surface is the same along any direction parallel to the centerline.

(9) In one example, the main body includes a first cross-section in a plane perpendicular to the first axis, and the inner and outer edges of the first cross-section are parallel arcs.

(10) In one example, the main body includes a first cross-section in a plane perpendicular to the first axis, the main body has an equivalent load region corresponding to the cross-section when the cutting blade is driven to rotate, and a ratio of an area of the first cross-section to an area of the equivalent load region is configured to be greater than or equal to 75%.

(11) In one example, the first cross-section substantially coincides with the equivalent load region.

(12) In one example, the main body includes a light-loaded section, a ratio of an area of any cross-section of the light-loaded section in a plane perpendicular to the first axis to an area of an equivalent load region of the any cross-section is configured to be greater than or equal to 75%, and a ratio of a length of the main body in a direction of a centerline of the main body to a length of the light-loaded section in the direction of the centerline of the main body is greater than or equal to 1 and less than or equal to 3.

(13) In one example, any cross-section of the light-loaded section in a plane perpendicular to the first axis coincides with an equivalent load region corresponding to the any cross-section.

(14) In one example, a cutting blade is capable of being driven to rotate around a first axis. The cutting blade includes cutting portions including cutting edges for cutting and extension portions closer to the first axis relative to the cutting portions. Each of the extension portions includes an upper surface and a lower surface, at least part of the upper surface is a part of the side surface of a first cone with the first axis as an axis, or at least part of the lower surface is a part of the side

surface of a second cone with the first axis as an axis.

(15) In one example, each of the extension portions includes a main body, the main body constitutes a main part of the each of the extension portions and includes a first section and a second section disposed on two sides of a centerline of the main body, and a vertical distance between each point, on an inner edge of a cross-section perpendicular to the first axis, of at least part of the first section and the first axis is the same.

(16) In one example, a vertical distance between each point, on an edge of a cross-section perpendicular to the first axis, of at least part of the second section and the first axis is the same.

(17) In one example, the upper surface is a part of the side surface of the first cone with the first axis as the axis.

(18) In one example, the lower surface is a part of the side surface of the second cone with the first axis as the axis.

(19) In one example, each of the extension portions includes a main body and a rib, the main body constitutes a main part of the each of the extension portions, and the rib is disposed on the main body.

(20) In one example, the main body includes a centerline, and a distance between the upper surface and the lower surface is the same along any direction parallel to the centerline.

(21) In one example, the main body includes a first cross-section in a plane perpendicular to the first axis, and the inner and outer edges of the first cross-section are parallel arcs.

(22) In one example, at least part of the upper surface is a part of the side surface of the first cone with the first axis as an axis, and at least part of the lower surface is a part of the side surface of the second cone with the first axis as an axis.

(23) In one example, a lawn mower includes a cutting blade, a base connected to the cutting blade, a motor configured to drive the cutting blade to rotate around a first axis and driving wheels configured to drive the base to move. The cutting blade includes cutting portions for cutting and extension portions closer to the first axis relative to the cutting portions, each of the extension portions includes an upper surface and a lower surface, at least part of the upper surface is a part of the side surface of a first cone with the first axis as an axis, or at least part of the lower surface is a part of the side surface of a second cone with the first axis as an axis.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a structural view of a lawn mower according to an example of the present disclosure;

(2) FIG. 2 is a perspective view of a cutting blade according to an example of the present disclosure;

(3) FIG. 3 is a plan view of the cutting blade in FIG. 2 of the present disclosure;

(4) FIG. 4 is a plan view of the cutting blade in FIG. 2 of the present disclosure from another angle;

(5) FIG. 5 is a sectional view of an extension portion of the cutting blade in FIG. 2 of the present disclosure;

(6) FIG. 6A is an equivalent load ring of the main body corresponding to a first section;

(7) FIG. 6B is an enlarged view of a partial area of FIG. 6A;

(8) FIG. 7 is a schematic view of a cross-section of the extension portion in FIG. 5 of the present disclosure and an equivalent load region of the cross-section;

(9) FIGS. 8A and 8B are schematic views of planes passing through a first axis according to an example of the present disclosure;

(10) FIG. 8C is a sectional view of an extension portion in the plane in FIG. 8A of the present disclosure;

(11) FIGS. 9A and 9B are schematic views illustrating a vertical distance between a point on an

edge of a cross-section of the extension portion in FIG. 5 and a first axis according to the present disclosure;

(12) FIG. 10 is a schematic view of a first cone and a second cone in the cutting blade in FIG. 2 of the present disclosure;

(13) FIG. 11 is a structural view of an existing second cutting blade;

(14) FIG. 12 is a sectional circular view of a cross-section of a second extension portion of the second cutting blade in FIG. 11 of the present disclosure;

(15) FIG. 13 is a comparison view of equivalent load regions of a first cutting blade and a second cutting blade;

(16) FIG. 14 is a schematic view of an equivalent load ring when a cross-section of the cutting blade in FIG. 5 rotates;

(17) FIG. 15 is a schematic view of an equivalent load ring when a cross-section of a second cutting blade rotates;

(18) FIG. 16 is a structural view of a light-loaded section of a cutting blade according to an example of the present disclosure;

(19) FIG. 17 is a structural view of a cutting blade according to an example of the present disclosure; and

(20) FIG. 18 is a sectional view of the cutting blade in FIG. 17.

DETAILED DESCRIPTION

(21) Examples in accordance with the present disclosure are described in detail hereinafter.

Examples are illustrated in the drawings, where the same or similar reference numerals indicate the same or similar elements or elements having the same or similar functions. The examples are intended to explain the present disclosure and are not to be construed as limiting the present disclosure.

(22) Technical solutions of the present disclosure are further described hereinafter in conjunction with drawings and examples.

(23) Referring to FIGS. 1 and 2, the present disclosure provides a cutting blade **100** and a lawn mower **200** using the cutting blade **100**. When the cutting blade **100** is mounted to the lawn mower **200**, the cutting blade **100** can be driven by the lawn mower **200** to rotate around a first axis **102**. The lawn mower **200** implements a mowing operation through the cutting blade **100**. The lawn mower **200** includes a base **210**, a motor **230**, a power supply, and driving wheels **220**. The cutting blade **100** is connected to the base **210**. The motor **230** is disposed in the base **210**. The blade and the base **210** are connected through the motor **230**. The motor **230** is configured to drive the cutting blade **100** to rotate. The cutting blade **100** is generally disposed in a chassis below the base **210** and cuts vegetation by rotating. The power supply is connected to the base **210** and configured to supply power to the motor **230**. The driving wheels **220** are connected to the base **210** to achieve the movement of the lawn mower **200**. Optionally, a user may supply power to drive the driving wheels **220** to move, or the motor **230** may be provided to supply power to the driving wheels **220** so that the driving wheels **220** is self-propelled.

(24) Referring to FIGS. 3 and 4, the cutting blade **100** includes a central portion **110**, cutting portions **120**, and extension portions **130**, where the central portion **110** is configured to be connected to the motor **230** of the lawn mower **200**. The lawn mower **200** includes an output shaft connected to the motor **230**, and the cutting blade **100** is detachably connected to the output shaft. The output shaft extends from an inside of the base **210** to a lower side of the base **210** to drive the cutting blade **100** to rotate. The central portion **110** is connected to the output shaft, and a cutting blade is driven to rotate through the output shaft when the motor **230** is operating. A rotation center **101** of the cutting blade **100** is disposed on the central portion **110**, and the cutting blade **100** can rotate around the first axis **102** passing through the rotation center **101**. The output shaft is connected to the cutting blade **100** at the rotation center **101** of the cutting blade **100**, and the motor **230** and the output shaft extend along the first axis **102** to drive the cutting blade **100** to rotate

around the first axis **102**. Optionally, the cutting blade **100** is symmetrical about the rotation center **101**, and the cutting portions **120** are disposed on two sides of the rotation center **101**. The extension portion **130** includes a main body **131** and a rib **132**. The extension portion **130** includes a centerline **103** passing through a center of a main body **131**. Optionally, the main body **131** is basically symmetrical about the centerline **103**. The centerline **103** is located substantially in a middle of the main body **131**. Referring to FIG. 3, a projection of the extension portion **130** along an axial direction of the first axis **102** extends along the centerline **103**.

(25) Optionally, the centerline **103** is perpendicular to the first axis **102**. In this manner, an extending direction of the cutting portion **120** is a radial direction of the first axis **102**. The cutting portions **120** include cutting edges **121** for cutting. The extension portion **130** is connected to the cutting portion **120** and the central portion **110**. The extension portions **130** are disposed on two sides of the rotation center **101**. The extension portions are closer to the first axis **102** relative to the cutting portions **120** and extend along the centerline **103** to two sides of the rotation center **101**. The extension portion **130** is disposed between the central portion **110** and the cutting portion **120**. Along the centerline **103** in a direction farther from the rotation center **101**, the extension portions **130** extend downward so that in the axial direction of the first axis **102**, a height of the cutting portion **120** is lower than a height of the central portion **110**. In this manner, the cutting portions **120** can cut vegetation at low places and the collision of obstacles such as the ground on the central portion **110** can be reduced.

(26) The rib **132** is disposed on the main body **131** and configured to strengthen the strength of the main body **131**. The main body **131** constitutes a main part of the extension portion **130**. The main body **131** referred to in the present disclosure should be regarded as a structure where the rib **132** is removed. For example, a convex part of the rib **132** is removed, and a recess generated by the rib **132** is filled so that a surface of the main body is smooth. An additional structure such as a convex on a surface of the extension portion **130** is not a part of the main body **131**.

(27) Optionally, the cutting edge is disposed at a side end of the extension portion and configured to cut vegetation.

(28) When the lawn mower **200** is in a mowing operation, an operation load of the cutting blade **100** is mainly composed of a cutting load generated by cutting vegetation and a blade load generated by the blade when the blade rotates. Therefore, reducing the blade load can effectively improve the endurance of the lawn mower **200** and improve the actual cutting load capacity of the blade. When the cutting blade **100** is driven to rotate, the main body **131** has an equivalent load region **1312** on any first cross-section **1311** in a plane perpendicular to of the first axis **102**. A ratio of an area of the first cross-section **1311** to an area of the equivalent load region **1312** of the first cross-section **1311** is configured to be greater than or equal to 75%. Optionally, the ratio of the area of the first cross-section **1311** to the area of the equivalent load region **1312** of the first cross-section **1311** is configured to be greater than or equal to 85%. Here, referring to FIGS. 5 to 7, the first cross-section **1311** of the main body **131** refers to a figure in which the cross-section of the extension portion **130** in the plane removes the cross-section of the rib **132** and fills the recess corresponding to the rib **132**. A point of the first cross-section **1311** that is farthest from the rotation center **101** is a farthest point P1, and a circle centered on a point on the rotation center **101** and passing through the farthest point P1 is the first equivalent load circle. A point of the first cross section **1311** that is nearest from the rotation center **101** is the nearest point P2, and a circle centered on the point on the rotation center **101** and passing through the nearest point P2 is the second equivalent load circle. A ring between the first equivalent load circle and the second equivalent load circle is an equivalent load ring **105**. A section on the equivalent load ring **105** passing through the outermost edge of the first cross-section **1311** is the equivalent load region **1312** of the first cross-section **1311**. The area of the equivalent load region **1312** of the first cross-section **1311** affects the load of the first-section **1311** of the cutting blade **100** when the cutting blade **100** is operating, that is, the larger the area of the equivalent load region **1312** is, the larger

the blade load generated when the cutting blade **100** rotates is. To ensure the strength of the cutting blade **100**, a dimension of the cutting blade **100** cannot be too small. Therefore, a difference between the area of the first cross-section **1311** of the cutting blade **100** and the area of the equivalent load region **1312** corresponding to the first cross-section **1311** is reduced so that while the strength of the cutting blade **100** is ensured, the area of the equivalent load region **1312** is reduced, the blade load when the cutting blade **100** rotates is reduced, and the torque generated when the blade rotates is effectively reduced. Here, the blade load may be understood as a load generated by driving the cutting blade **100** when the lawn mower **200** is unloaded.

(29) Referring to FIG. 5, in a direction along a radial direction of the centerline **103** and farther from the centerline **103**, the extension portion **130** has a downward trend (in the figure, a direction perpendicular to a plane where the figure is located and extending to a rear of the plane is a downward direction). In practice, the downward direction refers to a position of the driving wheel relative to the base, and a cutting portion is located relatively below the central portion. Referring to FIGS. 8A, 8B, and 8C, at least three planes **106** passing through the first axis **102** exist, and cross-sections of the main body **131** on the planes **106** are basically the same. Here, the cross-sections are the same, which means that shapes and dimensions of the cross-sections are the same; the cross-sections are basically the same, which means that extending shapes of the cross-sections are basically the same, and a difference between areas of the cross-sections is less than 10% of the area. The difference between the cross-sections caused by the deformation of the rib or the surface of the blade belongs to the case where the cross-sections are basically the same. For a conventional cutting blade, the case where the cross-sections crossed by two planes with a small included angle are basically the same does not belong to the “basically the same” as provided in the present disclosure. Optionally, at least three planes **106** passing through the first axis **102** exist, an angle between two of the at least three planes **106** is greater than or equal to 2 degrees, the cross-sections of the main body **131** on the planes **106** are basically the same, and the cross-sections are defined as second cross-sections **107**. In this manner, a difference between the area of the first cross-section **1311** of the cutting blade **100** and the area of the equivalent load region **1312** corresponding to the first cross-section **1311** can be effectively reduced. Here, the planes **106** pass through the first axis **102**, which means that the first axis **102** is located in the planes **106** and is a line of intersection of the three planes **106**. The main body **131** provided in the present disclosure should be understood as the main part of the extension portion **130**. The deformation of the surface of the extension portion **130**, or the addition or deletion of other structures should not be regarded as the main body of the extension portion provided in the present disclosure.

(30) Optionally, it is defined that the extension portion **130** and the cutting portion **120** have a connecting line **108**, the extension portion **130** and the cutting portion **120** are connected at the connecting line **108**, and a cross-section of the main body on any plane passing through the first axis **102** and cutting the connecting line **108** is the same.

(31) Referring to FIGS. 3, 9A and 9B, the main body **131** includes a first section **133** and a second section **134** disposed on two sides of the rib **132**. Optionally, the rib **132** passes through the centerline **103**, and the first section **133** and the second section **134** are symmetrical about the centerline **103**. A vertical distance L1 between each point, on an inner edge **1313** of the first cross-section **1311** perpendicular to the first axis **102**, of at least part of the first section **133** and the first axis **102** is the same. A vertical distance L1 between each point, on an edge of the first cross-section **1311** perpendicular to the first axis **102**, of at least part of the second section **134** and the first axis **102** is the same. Here, the inner edge **1313** of the first cross-section **1311** refers to an edge line of the first section **133** closer to the rotation center **101**, and the first cross-section **1311** should further include an outer edge line farther from the rotation center **101**, where a vertical distance between each point on the outer edge line and the first axis **102** is also equal. The vertical distance between each point on the inner edge **1313** of the first cross-section **1311** and the first axis **102** refers to a length of a vertical line of each point on the inner edge **1313** of the first cross-section

1311 to the first axis. The structure of the main body **131** is adapted to the rotational movement of the cutting blade so that the edge of the first cross-section **1311** of the main body **131** is consistent with an overall circular trajectory of the first cross-section **1311** when the lawn mower **200** rotates. In this manner, the space of the equivalent load region **1312** corresponding to the first cross-section **1311** is fully used so that while the overall strength of the cutting blade **100** is not reduced, the blade load of the cutting blade **100** is reduced.

(32) In the case where the vertical distance between each point, on the edge of the first cross-section **1311** perpendicular to the first axis **102**, of at least part of the first section **133** and the first axis **102** is the same, a cutting surface formed by the first cross-section **1311**, perpendicular to the first axis **102**, of at least part of the first section **133** when the cutting blade **100** rotates around the first axis **102** is a surface of revolution with the first axis **102** as a rotation axis so that the cutting surface formed by the first cross-section **1311**, perpendicular to the first axis **102**, of the first section **133** when the cutting blade **100** rotates around the first axis **102** is a ring surface.

(33) Referring to FIG. **10**, the main body **131** includes an upper surface **135** and a lower surface **136**. The upper surface **135** is a part of the side surface of a first cone **104** with the first axis **102** as an axis, and the lower surface **136** is a part of the side surface of a second cone **105** with the first axis **102** as an axis. Therefore, edges of the first cross-section **1311** of the first section **133** in the radial direction of the first axis **102** should be a line segment on the upper surface **135** and a line segment on the lower surface **136**, respectively. The edge of the first cross-section **1311** is an arc on a circle with a certain point on the first axis **102** as a center. Similarly, edges of the first cross-section **1311** of the second section **134** in the radial direction of the first axis **102** should be a line segment on the upper surface **135** and a line segment on the lower surface **136**, respectively. The edge of the first cross-section **1311** is an arc on a circle with a certain point on the first axis **102** as a center. Optionally, along any direction parallel to the centerline **103**, a distance between the upper surface **135** and the lower surface **136** is the same, and the inner and outer edges (**1311c**, **1311d**) of the first cross-section **1311** are parallel arcs, which improves the balance of the cutting blade.

(34) Optionally, the main body **131** is configured to have at least one first cross-section **1311** in the radial direction of the first axis **102**, and the first cross-section **1311** coincides with the equivalent load region **1312** so that the blade load of the cutting blade **100** can be reduced. Optionally, the main body **131** is configured to have at least one region; in this region, any first cross-section **1311** in the radial direction of the first axis **102** coincides with the corresponding equivalent load region **1312**, thereby improving the performance of the lawn mower **200**. Optionally, the cutting blade **100** is configured such that any first cross-section **1311** of the extension portion **130** of the cutting blade **100** in the radial direction of the first axis **102** coincides with the corresponding equivalent load region **1312**.

(35) Referring to FIG. **11**, the principle of reducing the blade load of the cutting blade **100** provided in the present disclosure will now be introduced in conjunction with a structure of the existing cutting blade **100**. To distinguish between cutting blades, the cutting blade **100** provided in the present disclosure is a first cutting blade, and the existing cutting blade is a second cutting blade **100a**. Referring to FIGS. **12** and **13**, in a second extension portion of the second cutting blade **100a**, in the radial direction of the first axis, a first cross-section **1311a** of the second extension portion is generally rectangular or irregular (not considering a structure of a rib). For example, the first cross-section **1311a** is a rectangle. An equivalent load region **1312a** corresponding to the first cross-section **1311a** is an arc passing through four points on an outermost edge of the first cross-section **1311a**, that is, in the second cutting blade **100a**, an area of the first cross-section **1311a** is less than an area of the corresponding equivalent load region **1312a**. In this case, since the blade is generally designed symmetrically about the centerline, cross-sections of the main body on at most two planes, passing through the first axis, of the blade are the same. Referring to FIGS. **14** and **15**, an area of an equivalent load ring **105a** when the second cutting blade is operating is greater than an area of the equivalent load ring **105** when the first cutting blade is operating. Therefore, a blade

load of the second cutting blade **100a** in operation is greater than the blade load of the first cutting blade in operation. According to experimental tests, the first cutting blade provided in the present disclosure can reduce the blade load by 5% to 20% compared to the traditional second cutting blade, thereby improving the time of endurance of the lawn mower and improving the mowing performance of the lawn mower.

(36) The main body **131** extends along the centerline **103** to two sides of the rotation center **101**. Referring to FIG. **16**, the main body **131** includes a light-loaded section **137**. A ratio of an area of any first cross-section **1311** of the light-loaded section **137** in the radial direction of the first axis **102** to an area of the equivalent load region **1312** of the first cross-section **1311** is configured to be greater than or equal to 75%. Optionally, the ratio of the area of the first cross-section **1311** to the area of the equivalent load region **1312** of the first cross-section **1311** is configured to be greater than or equal to 85%. A ratio of a length of the main body **131** in a direction of the centerline **103** to a length of the light-loaded section **137** in the direction of the centerline **103** is greater than or equal to 1 and less than or equal to 3, so as to improve the endurance of the lawn mower **200** and improve the cutting performance of the lawn mower **200**.

(37) Optionally, any first cross-section **1311** of the light-loaded section **137** in the radial direction of the first axis **102** coincides with the equivalent load region **1312** corresponding to the first cross-section **1311**.

(38) Optionally, in the case where the cutting blade **100** is driven to rotate, the cutting portion **120** has the equivalent load region **1312** on any first cross-section **1311** in the radial direction of the first axis **102**, the cutting portion **120** is configured to have at least one cross-section in the radial direction of the first axis **102**, and a ratio of an area of the cross-section to an area of the equivalent load region of the section is configured to be greater than or equal to 75%.

(39) Optionally, the light-loaded section is configured to be a section on the cutting blade, which may be a section in an extending direction of the centerline; along the extending direction of the centerline, the extension portion has a part that is not the light-loaded section. Optionally, the light-loaded section is configured to be a section on the cutting blade, which may be a section in the radial direction of the centerline; in the radial direction of the centerline, the extension portion has the light-loaded section and a non-light-loaded section. Therefore, at least part of the upper surface of the main body is a part of the side surface of a first cone with the first axis as the axis, and at least part of the lower surface of the main body is a part of the side surface of a second cone with the first axis as the axis.

(40) Optionally, referring to FIGS. **17** and **18**, an extension portion **130b** only includes a main body **131b** without a rib, and the structure of the main body **131b** is also adapted to the description in the preceding examples.

(41) Optionally, the extension portion further includes a rib, and cross-sections of the rib on at least three planes passing through the first axis are the same, that is, the structure of the rib is made similar to the structure of the main body, so as to reduce the blade load generated by the rib when the blade rotates.

(42) The preceding examples illustrate only the basic principles and features of the present disclosure. The present disclosure is not limited by the preceding examples. Various modifications and variations made without departing from the spirit and scope of the present disclosure fall within the scope of the present disclosure. The scope of the present disclosure is defined by the appended claims and their equivalent.

Claims

1. A cutting blade capable of being mounted to a lawn mower, and when the cutting blade is mounted to the lawn mower, the cutting blade is capable of being driven to rotate around a first axis, the cutting blade comprising: a central portion connectable to the lawn mower about the first

axis; cutting portions comprising cutting edges for cutting; and extension portions, coupling the central portion to the cutting portions, wherein the extension portions are closer to the first axis relative to the cutting portions, each of the extension portions comprises a main body, the main bodies constitute a main part of each of the extension portions, each of the extension portions is configured to have at least three planes passing through the first axis, and cross-sections of the main bodies on the at least three planes are substantially the same, and wherein transition lines formed between the central portion and the extension portions are an arcuate line extending an entire width of the cutting blade.

2. The cutting blade of claim 1, wherein each of the main bodies comprises a first section and a second section disposed on two sides of a centerline of each of the main bodies and a vertical distance between each point, on an inner edge of a cross-section perpendicular to the first axis, of at least part of the first section and the first axis is the same.

3. The cutting blade of claim 1, wherein each of the main bodies comprises an upper surface and a lower surface and the upper surface is a part of a side surface of a first cone with the first axis as an axis.

4. The cutting blade of claim 3, wherein the lower surface is a part of a side surface of a second cone with the first axis as an axis.

5. The cutting blade of claim 4, wherein each of the main bodies comprises a centerline and a distance between the upper surface and the lower surface is the same along any direction parallel to the centerline.

6. The cutting blade of claim 2, wherein each of the main bodies comprises a first cross-section in a plane perpendicular to the first axis and the inner and outer edges of the first cross-section are parallel arcs.

7. The cutting blade of claim 1, wherein each of the main bodies comprises a first cross-section in a plane perpendicular to the first axis, each of the main bodies has an equivalent load region corresponding to the cross-section when the cutting blade is driven to rotate, and a ratio of an area of the first cross-section to an area of the equivalent load region is configured to be greater than or equal to 75%.

8. The cutting blade of claim 7, wherein the first cross-section substantially coincides with the equivalent load region.

9. The cutting blade of claim 1, wherein each of the main bodies comprises a light-loaded section, a ratio of an area of any cross-section of the light-loaded section in a plane perpendicular to the first axis to an area of an equivalent load region of the any cross-section is configured to be greater than or equal to 75%, and a ratio of a length of the each of the main bodies in a direction of a centerline of each of the main bodies to a length of the light-loaded section in the direction of the centerline of each of the main bodies is greater than or equal to 1 and less than or equal to 3.

10. A cutting blade capable of being driven to rotate around a first axis, comprising: a central portion rotatable about the first axis; cutting portions comprising cutting edges for cutting; and extension portions, coupling the central portion to the cutting portions, wherein the extension portions are closer to the first axis relative to the cutting portions, each of the extension portions comprises an upper surface and a lower surface, and at least part of the upper surface is a part of a side surface of a first cone with the first axis as an axis of the first cone or at least part of the lower surface is a part of a side surface of a second cone with the first axis as an axis of the first cone, and wherein first transition lines formed between the central portion and the extension portions are an arcuate line extending an entire width of the cutting blade and second transition lines formed between the extension portions and the cutting portions is also an arcuate line extending the entire width of the cutting blade.

11. The cutting blade of claim 10, wherein each of the extension portions comprises a main body, each of the main bodies constitutes a main part of the each of the extension portions and comprises a first section and a second section disposed on two sides of a centerline of each of the main bodies,

and a vertical distance between each point, on an inner edge of a cross-section perpendicular to the first axis, of at least part of the first section and the first axis is the same.

12. The cutting blade of claim 11, wherein a vertical distance between each point, on an edge of a cross-section perpendicular to the first axis, of at least part of the second section and the first axis is the same.

13. The cutting blade of claim 10, wherein the upper surface is the part of the side surface of the first cone with the first axis as the axis of the first cone.

14. The cutting blade of claim 10, wherein the lower surface is the part of the side surface of the second cone with the first axis as the axis of the first cone.

15. The cutting blade of claim 10, wherein each of the extension portions further comprises a rib, and the rib is disposed on each of the main bodies.

16. The cutting blade of claim 15, wherein each of the main bodies comprises a centerline and a distance between the upper surface and the lower surface is the same along any direction parallel to the centerline.

17. The cutting blade of claim 15, wherein each of the main bodies comprises a first cross-section in a plane perpendicular to the first axis and the inner and outer edges of the first cross-section are parallel arcs.

18. The cutting blade of claim 10, wherein the at least the part of the upper surface is the part of the side surface of the first cone with the first axis as an axis of the first cone and the at least part of the lower surface is the part of the side surface of the second cone with the first axis as an axis of the first cone.

19. A lawn mower, comprising: a cutting blade; a base connected to the cutting blade; a motor configured to drive the cutting blade to rotate around a first axis; and driving wheels configured to drive the base to move; wherein the cutting blade comprises cutting portions for cutting and extension portions closer to the first axis relative to the cutting portions, each of the extension portions comprises an upper surface and a lower surface, and at least part of the upper surface is a part of a side surface of a first cone with the first axis as an axis of the first cone or at least part of the lower surface is a part of a side surface of a second cone with the first axis as an axis of the first cone, when the cutting blade is mounted to the lawn mower, the cutting blade is driven to rotate around a first axis, the cutting blade comprising: a central portion configured to be connected to the lawn mower; cutting portions comprising cutting edges for cutting; and extension portions, wherein the extension portions are closer to the first axis relative to the cutting portions, the extension portions are between the central portion and the cutting portions, a transition line is formed between the central portion and one of the extension portions, and the transition line is arcuate across an entirety of a width of the cutting blade.
